Week 7 Decision Journal – Model Choice

Context

A Logistic Regression model was developed during Week 6 to establish a baseline for emergency department triage prediction using the cleaned Yale EMMLC dataset.

During Week 7, a Random Forest classifier was implemented and benchmarked against the baseline to evaluate whether increased model complexity justified the additional computational cost.

Alternatives Considered:

1. Logistic Regression
    - Simple baseline model.
    - Highly interpretable.
    - Fast training and inference.
2. Random Forest
    - Ensemble learning algorithm capable of modelling non-linear relationships.
    - Produced higher overall predictive performance.
    - Moderate interpretability through feature importance.
3. Gradient Boosting / Neural Network
    - Considered as future alternatives.
    - Not implemented during this phase because Random Forest provided an appropriate comparison with the Week 6 baseline.

Decision

- Retain Logistic Regression as the recommended model for Phase 3 while continuing to investigate Random Forest for future deployment.

Reasoning

- Random Forest achieved higher overall accuracy, macro precision and macro F1-score than the Logistic Regression baseline.
- Logistic Regression trained faster, produced predictions more quickly and remained substantially easier for clinicians to interpret.
- Logistic Regression achieved higher recall for ESI Level 1 patients, making it more suitable for identifying critically ill patients in the current evaluation.

<u>Things I Do Not Know Yet</u>

- Whether additional hyperparameter tuning or class-balancing techniques could improve Random Forest performance on ESI Level 1 patients.
- How either model would perform when validated on emergency department data from Caribbean hospitals or during real-time clinical deployment.